ModelIt!

Student Activity Pack

Will Brain-Computer Interfaces Change Humanity?

Modeling Neural Signal Processing, Brain Plasticity, and Human-Machine Integration

Grade 11L3

NGSS HS-LS1-2, HS-LS1-3, HS-ETS1-1

Cause and Effect / Structure and Function

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **What do you already know about how the brain produces and transmits electrical signals, and how might we detect those signals from outside the skull?**

2. **If a paralyzed person could control a computer with their thoughts, what technology would need to exist and what challenges do you think it would face?**

3. **Draw a diagram showing how you think electrical signals from the brain could be captured, processed, and converted into commands for a robotic arm.**

4. **What is the difference between reading signals from the brain (detecting what someone intends) versus writing signals to the brain (creating sensations or experiences)?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

- Electrode Count and Density
- Signal-to-Noise Ratio
- Decoding Algorithm Accuracy
- Tissue Response
- Interface Invasiveness
- Brain Adaptation Rate
- Information Transfer Rate

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Motor Cortex Prosthetic Control

High-density intracortical array in motor cortex of paralyzed patient

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Noninvasive EEG Spelling System

Scalp EEG with 64 channels, no surgical implantation

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Long-Term Implant Degradation

Intracortical array tracked over 24 months with progressive tissue encapsulation

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Bidirectional Stimulation — The ability of the BCI to not only read neural signals but also write information back to the brain through electrical stimulation — enabling sensory feedback from prosthetic limbs, direct brain-to-brain communication, or therapeutic stimulation for conditions like depression and PTSD

Wireless Data Transmission — The system for transmitting neural data from implanted electrodes through the skull to external processors without wired connections — requiring miniaturized electronics, efficient power delivery through inductive charging, and data bandwidth sufficient for real-time processing of thousands of neural channels

Neural Privacy — The protection of neural data — thoughts, intentions, emotional states, memories — from unauthorized access, commercial exploitation, or government surveillance, raising unprecedented questions about mental privacy in an era when brain activity can be recorded, stored, decoded, and potentially manipulated

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Can we connect the human brain directly to computers — and if we can, should we fundamentally alter the boundary between human thought and machine processing? Use evidence from your simulation to support your answers.

1. **How does your model demonstrate the fundamental trade-off between signal quality and surgical invasiveness in brain-computer interface design?**

2. **Why is brain plasticity essential for BCI function, and what does this tell us about the relationship between biological and artificial intelligence?**

3. **What does your model reveal about why the tissue response is the primary barrier to long-term BCI performance?**

4. **How would your model change if electrode materials could be developed that the brain's immune system does not recognize as foreign?**

5. **What are the limitations of modeling brain-computer interfaces when individual brains differ enormously in neural organization and plasticity?**

8 STEM Challenge: Design a Brain-Computer Interface System

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YOUR ENGINEERING MISSION

Design a brain-computer interface system for a specific clinical application, addressing electrode technology, signal processing, decoding algorithms, biocompatibility, and the ethical framework for human testing.

THE SCENARIO

A neuroengineering startup has asked your team to design a BCI system for one of three clinical applications: (1) a motor prosthesis allowing quadriplegic patients to control a robotic arm, (2) a speech synthesis system for patients with locked-in syndrome, or (3) a seizure prediction and intervention device for severe epilepsy. Each application has different requirements for electrode placement, signal type, decoding speed, and long-term reliability. Your design must specify the recording technology, implantation strategy, decoding algorithm approach, biocompatibility solution, and the informed consent framework for clinical trials — because you are proposing to surgically implant electronic devices in human brains.

Engineering Design Thinking

1. HOW MANY ELECTRODES AND IN WHAT BRAIN REGION WOULD YOU PLACE THEM, AND WHAT IS YOUR RATIONALE BASED ON THE NEURAL SIGNALS YOUR APPLICATION REQUIRES?

2. WHAT DECODING ALGORITHM APPROACH WOULD YOU USE AND HOW WOULD YOU HANDLE THE CO-ADAPTATION BETWEEN THE USER'S BRAIN AND THE MACHINE LEARNING MODEL?

3. WHAT MATERIALS AND DESIGN FEATURES WOULD YOU INCORPORATE TO MINIMIZE THE TISSUE RESPONSE AND EXTEND THE FUNCTIONAL LIFETIME OF THE IMPLANT?

4. WHAT INFORMATION WOULD YOU INCLUDE IN THE INFORMED CONSENT DOCUMENT FOR THE FIRST CLINICAL TRIAL PATIENTS, GIVEN THAT LONG-TERM EFFECTS OF BRAIN IMPLANTS ARE NOT FULLY UNDERSTOOD?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Neuroengineers design and develop brain-computer interface hardware and software at companies like Neuralink, Blackrock Neurotech, and Synchron, earning $100,000-$200,000/year. Computational Neuroscientists develop the neural decoding algorithms that translate brain activity into digital commands, working at universities and tech companies, earning $90,000-$180,000/year. Neuroethicists specialize in the ethical, legal, and social implications of neurotechnology, advising companies, governments, and medical institutions on responsible development and deployment, earning $75,000-$150,000/year.